# MITCHELL CHATTERJEE

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## SUMMARY

Machine Learning Engineer experienced in building real-time, production-grade NLP, retrieval, and agentic systems. Skilled in foundation model development, from data pipelines to LLM orchestration and evaluation. Adept at bridging research and engineering to deliver scalable, high-impact AI solutions.

## EXPERIENCE

**Machine Learning Engineer** March 2026 – Present
Layer 6 AI, Toronto, Ontario — *TD Bank's AI research and development team building production ML systems.*

- Built the automated evaluation framework on the Agentic AI team, now used to evaluate agentic conversations across the entire bank.
- Cut deployment time for new automated evaluation solutions from roughly six months to a few weeks by standardizing the evaluation pipeline and making evaluators configurable.

**AI / Machine Learning Engineer** December 2024 – February 2026
JSI, Ottawa, Ontario — *Provider of digital intelligence solutions used by law enforcement and intelligence agencies.*

- Led the migration from a proprietary orchestration framework to Agno/LangChain, enabling a modular, LLM-based agentic retrieval and summarization system integrated with MCP; deployed on-premises across customer sites with 100+ GPUs.
- Built a real-time RAG pipeline to embed and retrieve billions of multimodal records (audio, text, image), significantly improving search latency and contextual relevance.
- Spearheaded development of the company-wide evaluation framework, leveraging DeepEval and A/B testing to rigorously measure KPIs, including relevance, latency, and interaction quality — enabling continuous feature improvement.

**Junior AI / Machine Learning Engineer** May 2019 – Sep 2022
JSI, Ottawa, Ontario

- Designed and deployed a real-time threat detection system that scanned high-volume multimodal data streams — including audio, text, images, and video — to detect and alert on high-risk activity using anomaly detection and proprietary ML models.
- Proposed and implemented a scalable, Kafka-based event-driven architecture for time-sensitive alerting workflows, reducing end-to-end notification latency by 33%.

## PUBLICATIONS

**Chatterjee, M.**, et al. "Toward Robust Automated Cardiovascular Arrhythmia Detection using Self-supervised Learning and 1D Vision Transformers." *Nature Scientific Reports*, 2026.

## EDUCATION

**Master of Computer Science (Thesis), Machine Learning** Sep 2022 – Sep 2024
Carleton University, Ottawa, ON

- GPA: 4.0 / 4.0 (Equivalent to 98.3%)
- Thesis: *Toward Robust Automated Cardiovascular Arrhythmia Detection using Self-supervised Learning and 1D Vision Transformers*
- **Published in *Nature Scientific Reports***, 2026.

- Used PEFT, SLURM, Ray Tune, and HPC infrastructure for large-scale self-supervised training and fine-tuning.
- Courses: Advanced Machine Learning, Natural Language Processing, Deep/Reinforcement Learning

**Honours Bachelor of Computer Science, AI Specialization** Sep 2016 – Dec 2020
University of Ottawa, Ottawa, ON

- GPA: 4.0 / 4.0 (Equivalent to 91%) — *summa cum laude*

## TECHNICAL STRENGTHS

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|---|---|---|
| **Machine Learning:** | Transformers / CNNs / RNNs | Time-Series / Healthcare Data |
| | PyTorch / NumPy / Pandas | NLP / Docker / Kubernetes |
| | HPC / RAG / FAISS | DeepEval / Agno / MCP |
| **Programming:** | Python / C# / JavaScript | SQL / Java / C/C++ |